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Analysis and Impacts of Negative Bias Temperature Instability (NBTI)

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***Abstract*— As the Integrated Circuits (IC) density keeps on increasing with the scaling of CMOS devices in each successive technology generation, reliability concerns mainly Negative Bias Temperature Instability (NBTI) becomes a major challenge. NBTI degrades the performance of a PMOS transistor under a negative gate stress. The after effects of NBTI include: (a) threshold voltage increase of PMOS transistor, (b) drain current degradation, and (c) speed degradation. Elevated temperature and the negative gate stress play an important role in degradation of Gate Oxide which further degrades the above said parameters. Before any circuit design Stress Analysis becomes important for any device in order to get the complete performance of the circuit. Negative bias temperature instability (NBTI) has become the dominant reliability concern for nanoscale PMOS transistors. In this paper basically we have analysed the effect of temperature variations on NBTI for a buffer.**

Keywords- NBTI, Reliability, Threshold Voltage, Temperature, AC DC Stress, EZwave.

I. INTRODUCTION

NBTI occurs under negative gate voltage (e.g., V_{gs}= -V_{DD}) and is measured as an increase in the magnitude of threshold voltage. It mostly affects the PMOS transistor [1] and degrades the device drive current, circuit speed, noise margin, and other factors. As the gate oxide gets thinner than 4nm, the threshold voltage change caused by NBTI for the PMOS transistor has become the dominant factor to limit the life time, which is much shorter than that defined by hot-carrier induced degradation (HCI) of the NMOS transistor. Furthermore, different from HCI that occurs only during dynamic switching, NBTI is caused during static stress on the oxide even without current flow. NBTI degradation in MOSFETs is explained by the reaction-diffusion model mentioned in next section.

A gradual shift of threshold voltage (V_T) over time is commonly observed in p type metal-oxide-semiconductor field-effect transistors (p-MOSFET or PMOS). This shift is mainly caused by: (1) voltage stress on the gate oxide (2) temperature, (3) the duty cycle of the stressing voltage (static stress as compared to dynamic stress). This effect has become more severe as: Transistor dimensions have shrunk, the electric field applied to the gate oxide has increased and the operating voltage has become lower. NBTI degrades the gate oxide by the interface states creation and hole trapping in the vicinity of the interface. An interface trap is created when a negative voltage is applied to the gate of a PMOS device for a prolonged time. An interface trap is located near the Si-oxide boundary where holes (positive charge) can get stuck, and hence, they shift the threshold voltage. NBTI is a result of continuous trap generation in Si-SiO2 interface of PMOS transistors. In bulk MOSFET structure, undesirable *Si* dangling bonds exist due to structural mismatch at the Si-SiO2 interface. These dangling bonds act as charged interfacial traps. Charge can flow between the semiconductor and the interface states. The net charge in these interface states is a function of the position of the fermi level in the bandgap. Hydrogen passivation is applied to the *Si* surface after the oxidation process to transform dangling *Si* atoms to *Si-H* bonds. However, with time, these *Si-H* bonds can easily break during operation (i.e., negative bias for PMOS). The broken bond acts as interfacial traps and increases the threshold voltage (V_T) of the device, thus affecting the performance of the IC. NBTI impact gets even worse in scaled technology due to the higher operation temperature and the usage of ultra thin oxide (i.e., higher oxide field).

The following conditions holds good for the NBTI effect

(1) PMOS needs to be inverted i.e. Formation of channel but doesn't need any current flow.

(2) Needs negative electric field across oxide layer (Enhanced at relatively high negative gate voltage Vgs).

(3) High temperature.

There are two components of NBTI based on the pre existing interface traps and the creation of the new interface states. They are classified as Permanent (Non-Recoverable) and temporary (Recoverable). Permanent NBTI is due to the new interface traps generation i.e. the electric field is able to break Si-H bonds located at the silicon-oxide interface. Temporary NBTI is due to some pre-existing traps present in the gate oxide. The pre existing traps get filled with the holes coming from the channel of PMOS. This is called temporary because the traps can be emptied when the stress voltage is removed. Even after the removal of the stress the final increase in threshold voltage is mainly due to the permanent NBTI and partly due to temporary NBTI as shown in the fig. 1 as Stress phase and Recovery phase.

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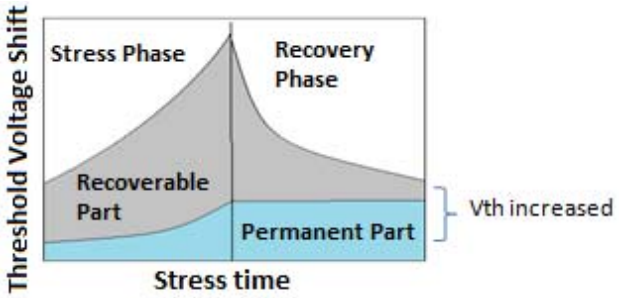

Figure 1: Temporary & Permanent phase of NBTI

II. Parameter Dependancies

As discussed above the main parameter which is affected by the NBTI is Threshold Voltage (V_T). NBTI raises threshold voltage above the initial value and hence subsequently degrades the other parameters like drain current, transconductance etc which depend on threshold voltage. The relationship between threshold voltage and interface trapped charges is given by

$$V_T = V_{FB} - 2\Phi_F - |Q_B| / C_{ox}$$

Where

$$\phi_F = (kT/q)\ln(N_D / n_i), |Q_B| = (4qK_S \in_0 \phi_F N_D)^{1/2}$$

$$V_{FB} = \phi_{MS} - \frac{Q_F}{C_{OX}} - \frac{Q_{it}(\phi_S)}{C_{OX}}$$

Where Q_F is the fixed charge density, Q_{it} is interface trap density C_{OX} is the gate oxide capacitance and Φ_{MS} is the work function between metal and semiconductor. The MOS drain current I_D (sat) and transconductance gm is related with threshold voltage as,

$$I_D = \left(W/2L\right)\mu_{eff} C_{OX} (V_G - V_T)^2$$

$$g_m = \left(W/2L\right)\mu_{eff} C_{OX} (V_G - V_T).$$

Thus we see that the Threshold voltage V_T of a MOS is dependent on Q_F and Q_{it}. As the threshold voltage is increased due to the NBTI (due to increase in interface traps) the drain current I_D and transconductance g_m also degrades.

A shift in the threshold voltage (V_T) ΔV_{th} of the PMOS transistor is proportional to the interface trap generation due to NBTI, which can be expressed as [2],

$$\Delta V_{th} = (1+m)\frac{qN_{it}(t)}{C_{OX}}$$

Where m represents equivalent V_T shifts due to mobility degradation (or model parameter), q is the electronic charge, and N_{it} (t) is the interface trap generation, which is the most important factor in evaluating performance degradation due to NBTI.

III. Mechanism of NBTI

Mechanism involved in Negative Bias Temperature Instability is better understood by the Reaction Diffusion model (RD model) [3], which explains the physics behind the degradation of the PMOS devices in terms of different sub-processes involving the bond breaking process and generation of interface traps.

According to the RD model, NBTI degradation originates from Silicon Hydrogen bonds (Si-H) breaking at Silicon-Silicon dioxide (Si-SiO2) interface during negative stress (V_{gs}=-V_{DD}), as shown in Figure 2. The broken Silicon bonds (Si-), dangling silicon act as interface traps that are responsible for higher V_T and lower drain current.

Interface traps Nit generation is due to the dissociation of Si-H bonds at the Si/SiO2 interface and subsequent movement of released hydrogen species away from the interface (diffusion), which leaves behind Si- dangling bonds (interface traps) .Inversion layer holes tunnel into the oxide and interact with Si-H bonds. The holes get captured and take away one electron from the Si-H bonds and make them weak. The weakened Si-H bonds then get broken by thermal excitation or otherwise. The released hydrogen species either diffuse away from the Si/SiO2 interface and leaves behind Si- (N_{it} generation), or reacts back with Si- and form Si-H (N_{it} passivation). It is worth noting that the magnitude of N_{it} is equal to the number of released H atoms at any given instant of time. The time evolution of N_{it} generation is modeled by the following equations.

$$Si - H \rightarrow Si^{\bullet} + H$$

$$Si - H + H^{+} \rightarrow Si^{\bullet} + H_2$$

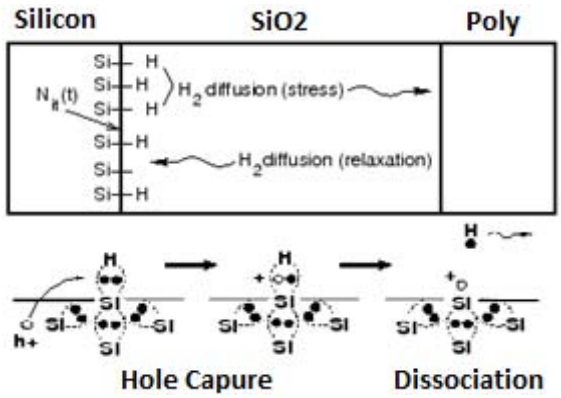

Figure 2: Representation of RD model [3]

The H atoms released from Si-H bond breaking contribute to three sub-processes including: (a) diffusion towards the gate, (b) combination with other H atoms to produce H2, or (c) recovery of the broken bonds. Similarly, H2 participate in the diffusion towards poly gate or dissociation to produce H atoms.

The R-D model takes the accumulation of hydrogen in the gate oxide as well as the loss of hydrogen into the poly into account in order to predict the long-time degradation. As the hydrogen molecules diffuses into the ploy from the SiO2, the probability of recovery (Si passivation) becomes very less so there is an increase in the interface trap generation and this is a temperature dominant process. As the temperature is increased the chances of diffusion of hydrogen molecules into the poly increases and hence decreases the probability of recovery during the recovery phase.

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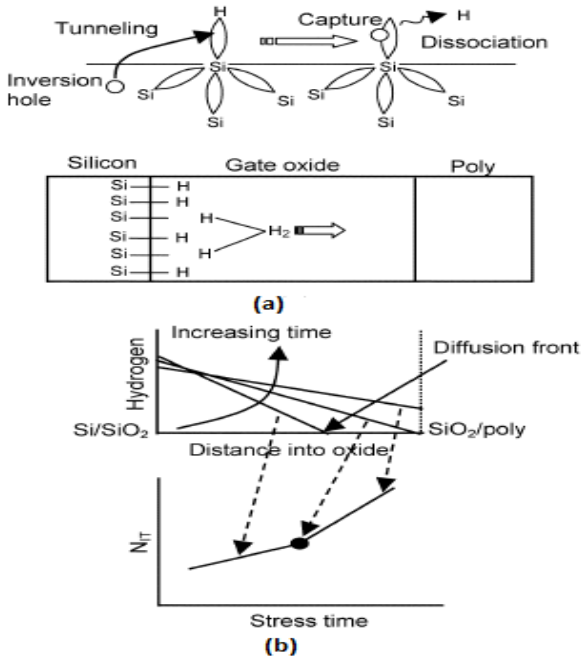

Figure 3: Fig showing RD model (a) Hole tunneling, capture and dissociation of Si-H bonds and subsequent diffusion of hydrogen away from the Si/SiO2 interface(b) Faster hydrogen diffusion after H front reaches the SiO2/poly interface, triggering faster interface-trap buildup [4]

The NBTI is mostly observed in the PMOS[1] devices only and appears to be negligible in NMOS because of the fact that interface traps can induce positive as well as negative charge at the interface. Interface traps readily exchange charge, either electrons or holes, with the substrate and they introduce either positive or negative net charge at interface, which depends on gate bias: the net charge in interface traps is negative in n-channel devices (NMOS), which are normally biased with positive gate voltage, but is positive in p-channel devices (PMOS) as they require negative gate bias to be turned on. On the other hand, charge found trapped in the centers in the oxide is generally positive in both n- and p-channel MOS transistors and cannot be quickly removed by altering the gate bias polarity. So the net effect i.e. interface charge and oxide charge is positive charge in case of PMOS and almost negligible for NMOS.

There can be two types of NBTI stress, it can be DC or AC stress. Once NBTI stress is removed from the device, a fraction of Interface traps N_{it} can self-anneal, resulting in V_{th} degradation being partially recovered. This recovery mechanism can be observed when a device is subject to a strain of stressing pulses. These conditions are called "AC" stress. Within the context of reliability, AC stress actually designates a large-signal pulse-like stress signal. During the first phase of the clock cycle, V_{th} increases due to the stress applied, and then it decreases again in the second half of the cycle when the stress is removed as shown in the figure 4 below. As shown in the figure, a CMOS inverter is drawn and when the V_g i.e. input gate voltage is zero (i.e. V_{gs}= -V_{DD}), the PMOS will be in the Stress phase and when V_g is V_{DD} (i.e. V_{gs}= 0) then PMOS will be in relaxation phase as shown. During Stress phase the effect of NBTI comes into picture.

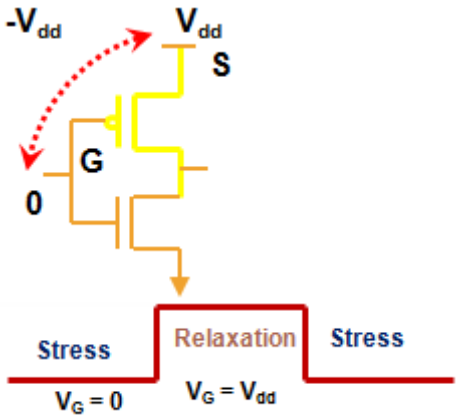

Figure 4: Pulse showing stress and relaxation phase of a PMOS

But the degradation rate is different (smaller) from "DC" stress conditions when the device is permanently stressed. This has often led to the conclusion that AC stress was less problematic than DC stress. The degradation rate under AC stress conditions actually depends on the duty cycle of the applied stress signal [5][7].

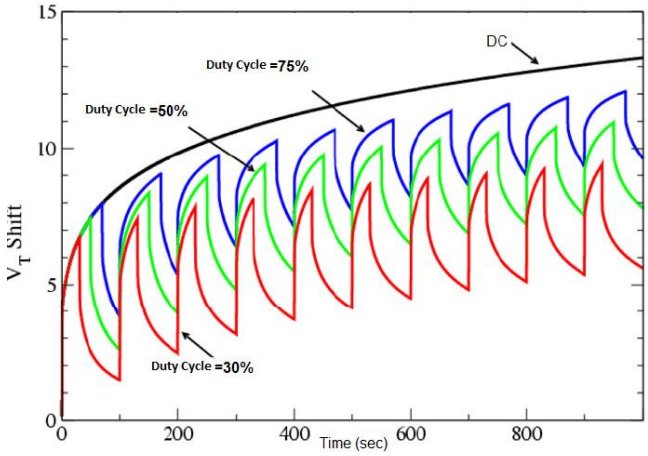

Figure 5: NBTI Degradation under DC and AC stress with different duty cycles [6]

IV. Simulation Results And Discussion

ELDO is a circuit simulator developed by Mentor Graphics, which delivers all the capability and accuracy of the SPICE level simulation for complex analog circuits and SoC designs. NBTI reliability simulation in Eldo is based on a model, which models the difference between the fresh and aged devices by calculating the NBTI stress which is dependent on the applied gate stress and the temperature. The following results are of a buffer circuit (for 45nm design), which has been simulated in the tool ELDO. The buffer is designed in a way such that two inverters are connected back to back and thus uses two PMOS (XIP1.M1 and XIP2.M1) and two NMOS. The below results are shown in the waveform viewer EZWave, used to view the output waveforms of ELDO file. Firstly a buffer circuit is described by a .cir file in the ELDO using the slow model of 45nm technology. Then simulation is carried out with this .cir file. During simulation, first stress file is generated in which stress for each device is calculated as per the ELDO's UDRM

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model and aged simulation uses this stress file to find the degradation.

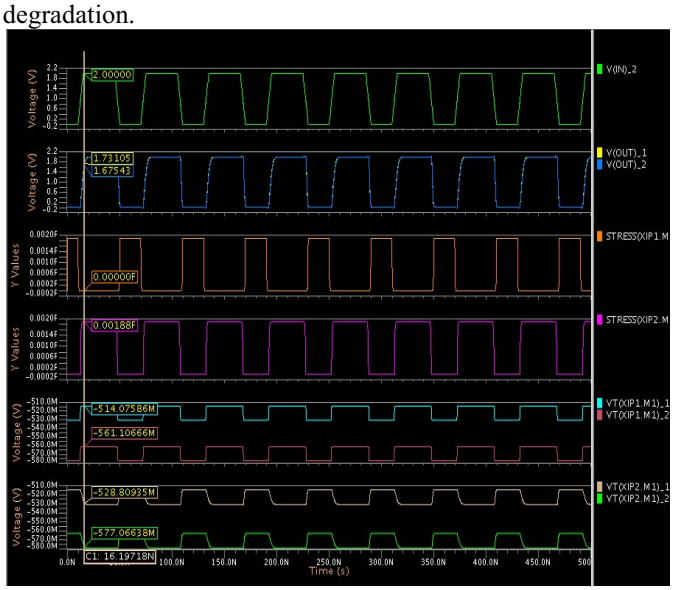

Figure 6: Buffer output showing input, output, stress and threshold voltage

The values shown are for the input pulse whose magnitude is 2V, rise time and fall time is 5nsec, pulse width is 30nsec and period is 60nsec. The transient analysis is done for 500ns in ELDO and the whole simulation is run for the period of 2years (i.e output is checked after 2 years). The V(OUT)_1 value in yellow color is fresh output and V(OUT)_2 with blue color is output after 2 years. At a particular time stamp of 16.19718ns the values are (shown in the rectangular boxes):

Type of Simulation	V(OUT) in V	VTH(XIP1.M1) in V	VTH(XIP2.M1) in V
Fresh	1.73105	-0.514075	-0.528809
After 2 yrs	1.67543	-0.561106	-0.577066

Apart from these values the instantaneous stress is also plotted for the two PMOS's as shown in the Fig 6. We can see the degradation in the output after 2 yrs due to stress in the PMOS devices.

The following result shows the buffer output at temperature 27'C

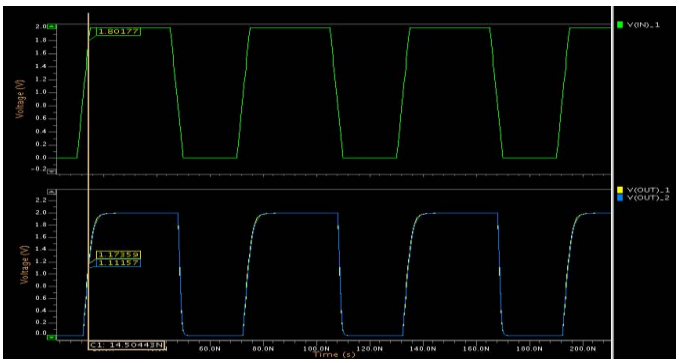

Figure 7: Input and Output of a buffer when temperature is 27'C

When the temperature is increased to 125 'C the output is,

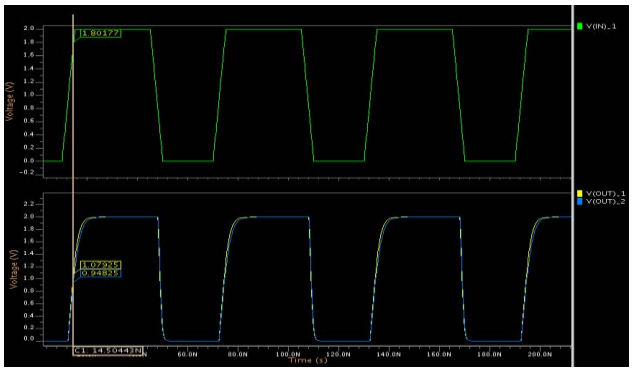

Figure 8:Input and Output of a buffer when temperature is 125'C

As can be seen from the results that when the temperature is increased from 27'C to 125'C the output is degraded much more (checked at the same time at t=14.504ns). For T=27'C V(OUT)_1 is 1.17359V and V(OUT)_2 is 1.11157, similarly for T=125'C V(OUT)_1 is 1.07925V and V(OUT)_2 is 0.94825V.

V. Conclusion

Based on the simulation results with an industrial 45nm technology, it is observed that the degradation of threshold voltage due to NBTI can be as high as 9% for a stress period of two years. As far as temperature variation is concerned, at room temperature the degradation of output is about 5% which is increased to about 11% at a temperature of 125'C.So Lower temperature is also desirable for robust nanoscale design. The transistor reliability will be a severe problem in future technology nodes which makes the device life time shorter than predicted.

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